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Quantum teleportation for continuous variables and related quantum information processing

Akira Furusawa[a,*], Nobuyuki Takei[b]

[a] *Department of Applied Physics, School of Engineering, The University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo 113-8656, Japan*
[b] *ERATO, Japan Science and Technology Agency, Japan*

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Abstract

Quantum teleportation is one of the most important subjects in quantum information science. This is because quantum teleportation can be regarded as not only quantum information transfer but also a building block for universal quantum information processing. Furthermore, deterministic quantum information processing is very important for efficient processing and it can be realized with continuous-variable quantum information processing. In this review, quantum teleportation for continuous variables and related quantum information processing are reviewed from these points of view.
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* Corresponding author.
E-mail address: akiraf@ap.t.u-tokyo.ac.jp (A. Furusawa)
URL: http://www.alice.t.u-tokyo.ac.jp/ (A. Furusawa).

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1. Basics of continuous-variable quantum information and quantum optics

Before explaining the details of quantum teleportation for continuous variables, we will briefly review the basics of continuous-variable (CV) quantum information and the related quantum optics.

There are two types of unit quantum states for quantum information. One is a quantum bit $|\psi^{(2)}\rangle$ (qubit), which is spanned by two orthogonal bases $|0\rangle$ and $|1\rangle$ and has the form of $|\psi^{(2)}\rangle = c_0|0\rangle + c_1|1\rangle = \sum_{n=0}^{1} c_n|n\rangle$ ($\langle i|j\rangle = \delta_{ij}$ $(i, j = 0, 1)$, $|c_0|^2 + |c_1|^2 = 1$). Here, the information is contained in c_0 and c_1. The other is a general quantum state $|\psi\rangle$, which is spanned by continuous bases $|x\rangle$ (x: a real number) and has the form of $|\psi\rangle = \int_{-\infty}^{\infty} \mathrm{d}x\psi(x)|x\rangle$ ($\langle x|x'\rangle = \delta(x - x')$, $\psi(x) = \langle x|\psi\rangle$). This is called CV quantum information. In this form, the information is contained in a wave function $\psi(x)$. Moreover, $|\psi\rangle$ can be expanded by discrete bases $|n\rangle$ $(n = 0, 1, 2, \ldots)$ like $|\psi\rangle = \sum_{n=0}^{\infty} c_n|n\rangle$ and then the information is contained in c_n. In this sense, a qubit is included in CV. We concentrate on CV quantum information processing in this review.

We use quantum optics for the realization of CV quantum information processing. In quantum optics, an optical field is quantized with annihilation and creation operators $\hat{a}$ and $\hat{a}^\dagger$, which are decomposed into quadrature-phase amplitude operators $\hat{x}$ and $\hat{p}$ in the form of $\hat{a} = \hat{x} + \mathrm{i}\hat{p}$ and $\hat{a}^\dagger = \hat{x} - \mathrm{i}\hat{p}$ with *photon-number unit* $\hbar = 1/2$. From the commutation relation of $\hat{a}$ and $\hat{a}^\dagger$, i.e., $[\hat{a}, \hat{a}^\dagger] = 1$, we can get $[\hat{x}, \hat{p}] = \mathrm{i}/2$ and it is the same commutation relationship of position and momentum which are canonically conjugate variables. Thus we can treat the quadrature-phase amplitude operators $\hat{x}$ and $\hat{p}$ as position and momentum.

For CV quantum information processing, quantum entanglement is one of the most important ingredients. With quantum optics, we can easily create it with squeezed states of light [1,2]. Fig. 1 shows one of the examples of creating quantum entanglement with squeezed vacua, and the operator description is shown in Fig. 2. Note that the same state can be directly created with so-called type-II phase matching of a second order nonlinear crystal [2].

The output state $|\mathrm{EPR}*\rangle_{\mathrm{AB}}$ of the beam splitter is

$$\begin{aligned} |\mathrm{EPR}*\rangle_{\mathrm{AB}} &= \hat{B}_{\mathrm{HBS}}\hat{S}_{\mathrm{A}}(-r)\hat{S}_{\mathrm{B}}(r)|0\rangle_{\mathrm{A}} \otimes |0\rangle_{\mathrm{B}} \\ &= \exp[r(\hat{a}_{\mathrm{A}}^\dagger\hat{a}_{\mathrm{B}}^\dagger - \hat{a}_{\mathrm{A}}\hat{a}_{\mathrm{B}})]|0\rangle_{\mathrm{A}} \otimes |0\rangle_{\mathrm{B}}, \end{aligned} \tag{1}$$

where a half-beam-splitter operator $\hat{B}_{\mathrm{HBS}}$ satisfies the following relation:

$$\hat{B}_{\mathrm{HBS}}^\dagger \begin{pmatrix} \hat{a}_{\mathrm{A}} \\ \hat{a}_{\mathrm{B}} \end{pmatrix} \hat{B}_{\mathrm{HBS}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \hat{a}_{\mathrm{A}} \\ \hat{a}_{\mathrm{B}} \end{pmatrix}, \tag{2}$$

and the definition of a squeezing operator is $\hat{S}(r) = \exp[\frac{r}{2}(\hat{a}^2 - \hat{a}^{\dagger 2})]$ and r is a squeezing parameter.

Eq. (1) can be simplified as follows [3]:

$$|\mathrm{EPR}*\rangle_{\mathrm{AB}} = \sqrt{1 - q^2} \sum_{n=0}^{\infty} q^n |n\rangle_{\mathrm{A}} \otimes |n\rangle_{\mathrm{B}}, \tag{3}$$

where $q = \tanh r$. From this form, it is obvious that modes A and B are entangled when $q > 0$. Moreover, when $r \to \infty$ so $q \to 1$, the state approaches the Einstein–Podolsky–Rosen (EPR) state, which is defined as a simultaneous eigenstate of two commuting operators $\hat{x}_{\mathrm{A}} - \hat{x}_{\mathrm{B}}$ and $\hat{p}_{\mathrm{A}} + \hat{p}_{\mathrm{B}}$ [4]. Note that the eigen values are $x_{\mathrm{A}} - x_{\mathrm{B}} = 0$ and $p_{\mathrm{A}} + p_{\mathrm{B}} = 0$ in the present case. The EPR state $|\mathrm{EPR}\rangle_{\mathrm{AB}}$ is represented with continuous bases $|x\rangle$ or $|p\rangle$ as

$$\begin{aligned} |\mathrm{EPR}\rangle_{\mathrm{AB}} &= c \int_{-\infty}^{\infty} \mathrm{d}x |x\rangle_{\mathrm{A}} \otimes |x\rangle_{\mathrm{B}} \\ &= c \int_{-\infty}^{\infty} \mathrm{d}p |p\rangle_{\mathrm{A}} \otimes |-p\rangle_{\mathrm{B}}. \end{aligned} \tag{4}$$

This type of EPR correlation is experimentally confirmed as shown in Fig. 3 [5]. The *EPR beams* are created with the scheme of Fig. 1. It is clear from the figure that x_{A} and x_{B} are correlated and p_{A} and p_{B} are anti-correlated. It follows that the created state is an approximate EPR state.

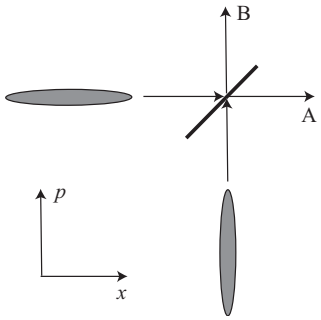


Fig. 1. Creation of quantum entanglement with squeezed vacua and a half beam splitter. Ellipses denote squeezed vacua.

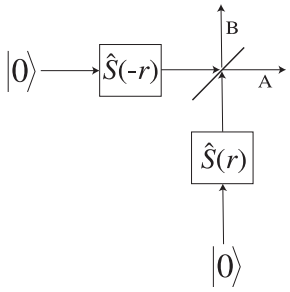


Fig. 2. Operator description of Fig. 1. $|0\rangle$ denotes an initial vacuum and $\hat{S}(r)$ denotes a squeezing operator.

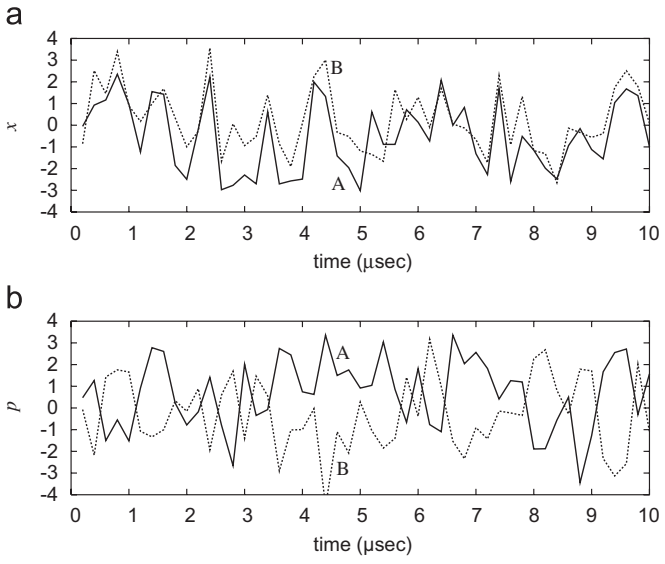


Fig. 3. Time domain EPR correlation created with two squeezed vacua (squeezing level is ~ -4 dB) and a half beam splitter [5]. In (a), A and B denote x_{A} and x_{B}, and in (b), A and B denote p_{A} and p_{B}, respectively.

Sufficient criteria for the existence of bipartite quantum entanglement between modes A and B are proposed. One of the most convenient criteria for experimentalists is following [6,7].

$$\Delta_{\mathrm{A,B}} \equiv \langle[\Delta(\hat{x}_{\mathrm{A}} - \hat{x}_{\mathrm{B}})]^2\rangle + \langle[\Delta(\hat{p}_{\mathrm{A}} + \hat{p}_{\mathrm{B}})]^2\rangle < 1. \tag{5}$$

Of course, the left-hand side of Eq. (5) is zero for the ideal EPR state. With this criterion, we can check existence of entanglement between modes A and B in the state $|\mathrm{EPR}*\rangle_{\mathrm{AB}}$ of Eq. (3). When $q>0$, the criterion is satisfied and there is entanglement between modes A and B.

2. CV quantum teleportation

2.1. Theoretical background of CV quantum teleportation

Quantum teleportation for qubit systems was originally proposed by Bennett et al., in 1993 [8]. Vaidman extended the concept to general quantum states, i.e., continuous variables [9]. Braunstein and Kimble proposed the experimental scheme with squeezed states [10]. Moreover, the scheme was extended to atomic systems [11–13]. We will briefly introduce the Braunstein–Kimble scheme here [14].

Fig. 4 shows the Braunstein–Kimble scheme of CV quantum teleportation. First Alice and Bob share the entangled state $|\mathrm{EPR}*\rangle_{\mathrm{AB}}$ created with two squeezed vacua. Then Alice makes joint measurement between modes V and A and gets the values of x_- and p_+, where mode V is an input for teleportation, mode A is one of the EPR beams, $\hat{x}_- = \hat{x}_{\mathrm{V}} - \hat{x}_{\mathrm{A}}$, and $\hat{p}_+ = \hat{p}_{\mathrm{V}} + \hat{p}_{\mathrm{A}}$. The simultaneous measurement of x_- and p_+ corresponds to the projection onto a simultaneous eigenstate of $\hat{x}_-$ and $\hat{p}_+$ [14]:

$$|x_-, p_+\rangle = \frac{1}{\pi}\sum_{n=0}^{\infty} \hat{D}_{\mathrm{V}}(x_- + \mathrm{i}p_+)|n\rangle_{\mathrm{V}} \otimes |n\rangle_{\mathrm{A}}, \tag{6}$$

where $\hat{D}_{\mathrm{V}}(x_- + \mathrm{i}p_+)$ is a displacement operator in phase space and the definition is $\hat{D}_{\mathrm{V}}(x_- + \mathrm{i}p_+) = \exp[2\mathrm{i}(p_+\hat{x} - x_-\hat{p})]$.

Before the joint measurement, the overall state is $|\psi_{\mathrm{in}}\rangle \otimes |\mathrm{EPR}*\rangle_{\mathrm{AB}}$. Just after the measurement, the overall state becomes an inner product of $\langle x_-, p_+|$ and $|\psi_{\mathrm{in}}\rangle \otimes |\mathrm{EPR}\rangle_{\mathrm{AB}}$, which is

$$|\Psi(x_-, p_+)\rangle_{\mathrm{B}} = \sqrt{\frac{1-q^2}{\pi}}\sum_{n=0}^{\infty} q^n |n\rangle\langle n|\hat{D}(-x_- - \mathrm{i}p_+)|\psi_{\mathrm{in}}\rangle_{\mathrm{B}}. \tag{7}$$

When $q \to 1$, $\sum_{n=0}^{\infty} q^n |n\rangle\langle n|$ approaches identity $\hat{I}$. In this case, the state at Bob $|\Psi(x_-, p_+)\rangle_{\mathrm{B}}$ becomes the displaced input state by $(-x_- - \mathrm{i}p_+)$ in phase space.

Alice sends the information of x_- and p_+ to Bob and he displaces his state by $(x_- + \mathrm{i}p_+)$ in phase space. When $q \to 1$, the state at Bob becomes the input state $|\psi_{\mathrm{in}}\rangle$. Of course, we cannot get $q \to 1$ because we cannot get infinite amount of squeezing. So, in the real experiment, the state at Bob is an approximate state to the input.

Evaluation of the performance of teleportation is conducted with fidelity F. For pure-state inputs, the fidelity is defined as $F = \langle\psi_{\mathrm{in}}|\hat{\rho}_{\mathrm{out}}|\psi_{\mathrm{in}}\rangle$, where $|\psi_{\mathrm{in}}\rangle$ is a state vector of an input and $\hat{\rho}_{\mathrm{out}}$ is a density operator of the output. The classical limit for a coherent state input is $F_{\mathrm{coh}} = 1/2$ [15,16]. So experimentalists try to get a better value of the fidelity than the classical limit to claim success of teleportation. Another benchmark of the performance of teleportation is the fidelity of 2/3 for a coherent state input. $F_{\mathrm{coh}} = 2/3$ is referred to as the *no cloning limit*, because surpassing this limit warrants that the teleported state is the best remaining copy of the input state [17]. Moreover, a teleporter surpassing

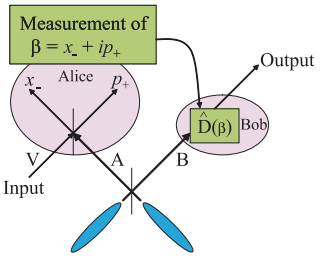


Fig. 4. Braunstein–Kimble scheme of CV quantum teleportation.

$F_{\rm coh} = 2/3$ enables the transfer of nonclassicality in an input state, where the nonclassicality means a negative part of Wigner function for a quantum state like the Schrödinger cat state $|\psi_{\rm cat}\rangle \propto |\alpha\rangle - |-\alpha\rangle$ and a single photon state [18]. Note that the fidelity F is an input-state dependent measure and we always have to take account of the input state [19].

For convenience of later use, we introduce the Heisenberg picture of CV quantum teleportation. First of all, Alice and Bob share the EPR beams A and B which are created with the scheme shown in Fig. 1. Squeezed vacua ($\hat{a}_1$, $\hat{a}_2$) are expressed as

$$\begin{aligned} \hat{a}_1 &= e^r \hat{x}_1^{(0)} + {\rm i} e^{-r} \hat{p}_1^{(0)}, \\ \hat{a}_2 &= e^{-r} \hat{x}_2^{(0)} + {\rm i} e^r \hat{p}_2^{(0)}, \end{aligned} \tag{8}$$

where $\hat{x}_1^{(0)}$, $\hat{p}_1^{(0)}$, $\hat{x}_2^{(0)}$, and $\hat{p}_2^{(0)}$ are quadrature-phase amplitudes of initial vacuum states, $\langle(\Delta\hat{x}_1^{(0)})^2\rangle = \langle(\Delta\hat{p}_1^{(0)})^2\rangle = \langle(\Delta\hat{x}_2^{(0)})^2\rangle = \langle(\Delta\hat{p}_2^{(0)})^2\rangle = 1/4$, and r is a squeezing parameter. By combining these two modes with a half beam splitter, the two output beams become the EPR beams. Output modes ($\hat{a}_{\rm A}$, $\hat{a}_{\rm B}$) from the half beam splitter are expressed as

$$\begin{aligned} \hat{a}_{\rm A} &= \frac{1}{\sqrt{2}}(\hat{a}_1 + \hat{a}_2), \\ \hat{a}_{\rm B} &= \frac{1}{\sqrt{2}}(\hat{a}_1 - \hat{a}_2). \end{aligned} \tag{9}$$

Thus the output modes ($\hat{a}_{\rm A} = \hat{x}_{\rm A} + {\rm i}\hat{p}_{\rm A}$, $\hat{a}_{\rm B} = \hat{x}_{\rm B} + {\rm i}\hat{p}_{\rm B}$) show the following correlation:

$$\begin{aligned} \hat{x}_{\rm A} - \hat{x}_{\rm B} &= \sqrt{2} e^{-r} \hat{x}_2^{(0)}, \\ \hat{p}_{\rm A} + \hat{p}_{\rm B} &= \sqrt{2} e^{-r} \hat{p}_1^{(0)}. \end{aligned} \tag{10}$$

It is obvious that modes A and B satisfy Eq. (5) and they are entangled when $r > 0$.

After combining an input and mode A with a half beam splitter, Alice measures $\hat{x}_- = (\hat{x}_{\rm in} - \hat{x}_{\rm A})/\sqrt{2}$ and $\hat{p}_+ = (\hat{p}_{\rm in} + \hat{p}_{\rm A})/\sqrt{2}$ as shown in Fig. 4. Bob's mode B before the measurement is $\hat{x}_{\rm B} = \hat{x}_{\rm in} - (\hat{x}_{\rm A} - \hat{x}_{\rm B}) - \sqrt{2}\hat{x}_-$ and $\hat{p}_{\rm B} = \hat{p}_{\rm in} + (\hat{p}_{\rm A} + \hat{p}_{\rm B}) - \sqrt{2}\hat{p}_+$. After the Alice's measurement, operators $\hat{x}_-$ and $\hat{p}_+$ collapse and reduce to certain values. Receiving her measurement results through classical channels, Bob displaces his mode as $\hat{x}_{\rm B} \to \hat{x}_{\rm out} = \hat{x}_{\rm B} + \sqrt{2} g_x x_-$ and $\hat{p}_{\rm B} \to \hat{p}_{\rm out} = \hat{p}_{\rm B} + \sqrt{2} g_p p_+$. Here g_x and g_p are information gains of classical channels from Alice to Bob and they are usually $g_x = g_p = 1$. Thus with Eq. (10) the teleported output mode becomes:

$$\begin{aligned} \hat{x}_{\rm out} &= \hat{x}_{\rm in} - (\hat{x}_{\rm A} - \hat{x}_{\rm B}) = \hat{x}_{\rm in} - \sqrt{2} e^{-r} \hat{x}_2^{(0)}, \\ \hat{p}_{\rm out} &= \hat{p}_{\rm in} + (\hat{p}_{\rm A} + \hat{p}_{\rm B}) = \hat{p}_{\rm in} - \sqrt{2} e^{-r} \hat{p}_1^{(0)}. \end{aligned} \tag{11}$$

For a coherent state input, the teleportation fidelity $F_{\rm coh}$ is calculated as [20]:

$$\begin{aligned} F_{\rm coh} &= \frac{2}{\sqrt{(1 + 4\langle(\Delta\hat{x}_{\rm out})^2\rangle)(1 + 4\langle(\Delta\hat{p}_{\rm out})^2\rangle)}} \\ &= \frac{1}{1 + e^{-2r}}. \end{aligned} \tag{12}$$

In the ideal case ($r \to \infty$), the teleportation is perfect ($F_{\rm coh} \to 1$). For the success of teleportation of a coherent state, it is sufficient to have $r > 0$ which gives $F_{\rm coh} > 1/2$. In other words, the success of teleportation means the existence of quantum entanglement between modes A and B.

2.2. CV quantum teleportation experiments

The Braunstein–Kimble scheme of CV quantum teleportation was experimentally realized by Furusawa et al. at Caltech in 1998 and the experimental setup is shown in Fig. 5 [20]. The input states were coherent states and the fidelity was $F_{\rm coh} = 0.58 \pm 0.02$ which is slightly better than the classical limit of $F_{\rm coh} = 1/2$. After the first realization, several

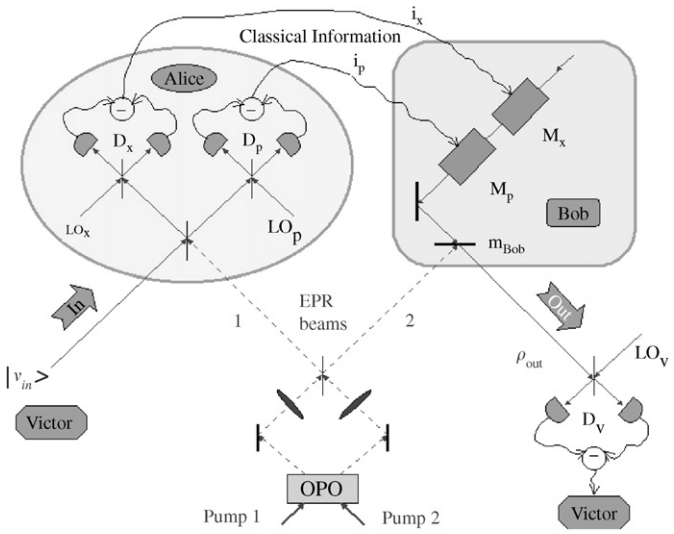


Fig. 5. Experimental realization of CV quantum teleportation [20]. Victor is a verifier who creates an input coherent state with a modulator and checks the output states with a homodyne detector. EPR beams are created with two squeezed vacua from an optical parametric oscillator (OPO) and a half beam splitter. Alice makes measurements of $\hat{x}_-$ and $\hat{p}_+$ with homodyne detectors whose local oscillator (LO) are set to x and p quadratures, respectively. With the information (electric signals) from Alice, Bob makes displacement operation with two modulators and a high reflectance mirror.

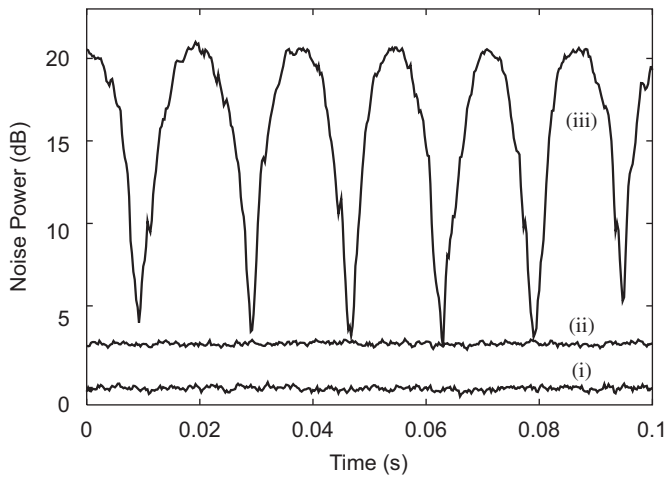


Fig. 6. The measurement results of the teleported state for a coherent state input in x quadrature [23]. Each trace is normalized to the corresponding vacuum noise level. (i) The corresponding vacuum noise level $\langle(\Delta\hat{x}_{\rm out}^{(0)})^2\rangle = 1/4$. (ii) The teleported state for a vacuum input. $\langle(\Delta\hat{x}_{\rm out})^2\rangle = 2.82 \pm 0.09$ dB and $\langle(\Delta\hat{p}_{\rm out})^2\rangle = 2.64 \pm 0.08$ dB (not shown). With these values, we get $F_{\rm coh} = 0.70 \pm 0.02$ by using Eq. (12). (iii) The teleported state for a coherent state input with the phase scanned. The measurement frequency is centered at 1 MHz, and the resolution and video bandwidths are 30 kHz and 300 Hz, respectively. Traces i and ii are averaged 20 times.

experimental realizations of CV quantum teleportation of optical fields have been reported [19,21–23]. The best fidelity so far for a coherent state input is $F_{\rm coh} = 0.70 \pm 0.02$ which was achieved by Takei et al. at Tokyo in 2005 [23]. The experimental result is shown in Fig. 6. In the experiment, squeezing level of the initial squeezed vacua was about -4 dB and it gives the fidelity of $F_{\rm coh} = 0.72$ from Eq. (12). The experimental fidelity of $F_{\rm coh} = 0.70 \pm 0.02$ almost agrees with the expected value and it follows that the experimental setup is almost ideal and the limiting factor for the fidelity is only the squeezing level of the resource of quantum entanglement. So we need more squeezing for getting further fidelity. Recently, Suzuki et al. succeeded to get -7 dB of squeezing which is supposed to give the fidelity of 0.83 [24].

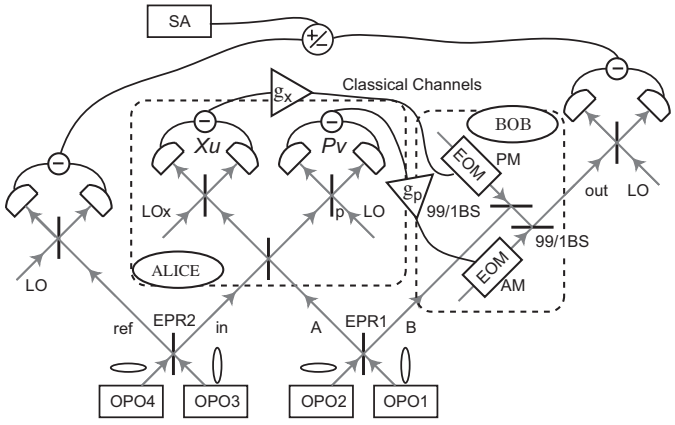


Fig. 7. The experimental setup for teleportation of quantum entanglement or entanglement swapping [23].

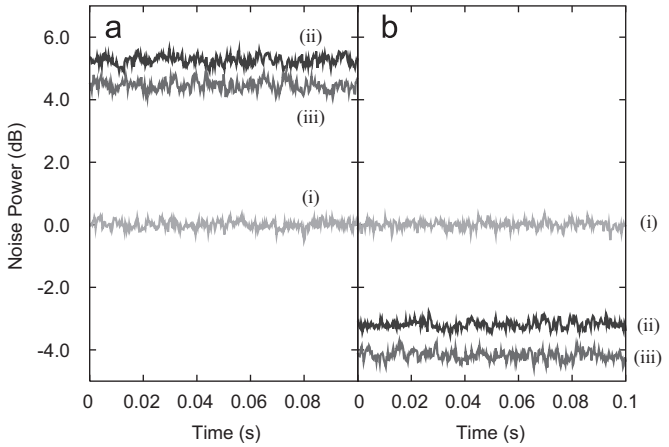


Fig. 8. Correlation measurement results for EPR2 beams. (a) The measurement results of the reference mode alone. (i) $\langle(\Delta\hat{x}_{\rm ref}^{(0)})^2\rangle = \langle(\Delta\hat{p}_{\rm ref}^{(0)})^2\rangle = 1/4$; (ii) $\langle(\Delta\hat{x}_{\rm ref})^2\rangle$; (iii) $\langle(\Delta\hat{p}_{\rm ref})^2\rangle$. (b) The measurement results of the correlation between the input and reference modes. (i) $\langle(\Delta\hat{x}_{\rm ref}^{(0)} - \Delta\hat{x}_{\rm in}^{(0)})^2\rangle = \langle(\Delta\hat{p}_{\rm ref}^{(0)} + \Delta\hat{p}_{\rm in}^{(0)})^2\rangle = 1/2$; (ii) $\langle(\Delta\hat{x}_{\rm ref} - \Delta\hat{x}_{\rm in})^2\rangle$; (iii) $\langle(\Delta\hat{p}_{\rm ref} + \Delta\hat{p}_{\rm in})^2\rangle$. The measurement conditions are the same as those of Fig. 6.

As pointed out in the previous subsection, the teleportation fidelity of $F_{\rm coh} = 0.70 \pm 0.02$ surpasses the no-cloning limit. It means that we can teleport nonclassicality of input states. As an example of teleportation of nonclassicality, we will show here teleportation of quantum entanglement or entanglement swapping. Fig. 7 shows the setup for teleportation of quantum entanglement.

Two pairs of entangled beams denoted by EPR1 and EPR2 are generated by combining two squeezed vacua at half beam splitters. The EPR1 beams consist of modes A and B, and they are utilized as a resource of teleportation. They shows the quantum entanglement of $\Delta_{\rm A,B} = 0.42 \pm 0.01 < 1$, where the definition of $\Delta_{\rm A,B}$ is in Eq. (5). It allows us to achieve the fidelity of 0.70 ± 0.01 for a coherent state input. On the other hand, the EPR2 beams consist of two subsystems: *reference* and *input*. While the reference is kept during a teleportation process, the input is teleported to an output station. After the process, success of this protocol is verified by examining quantum entanglement between the reference and the output: $\Delta_{\rm ref,out} < 1$.

Before performing the experiment of teleportation of quantum entanglement, the noise power of each mode for EPR2 beams and the initial correlation between the modes are measured with homodyne detectors and a spectrum analyzer (SA). The results are shown in Fig. 8. For the reference mode, the noise levels of 5.23 ± 0.14 and 4.44 ± 0.14 dB are

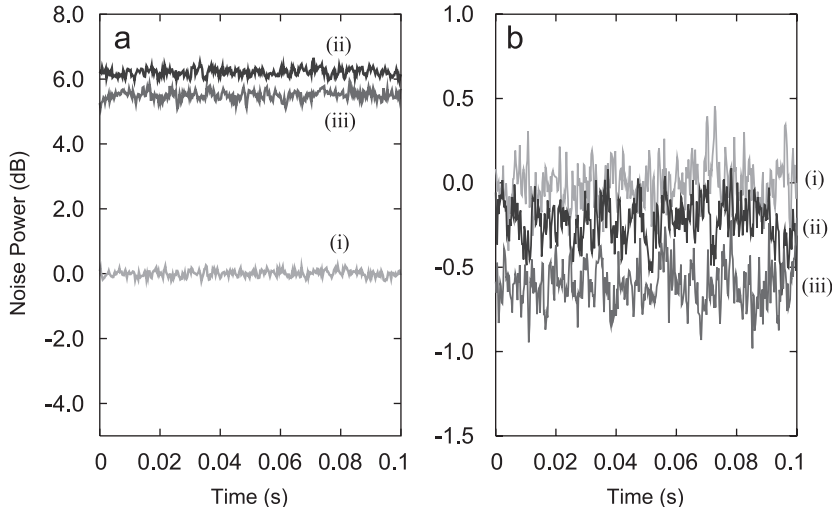


Fig. 9. Correlation measurement results for the output of teleportation of quantum entanglement. (a) The measurement results of the output mode alone. (i) $\langle(\Delta\hat{x}_{\rm out}^{(0)})^2\rangle = \langle(\Delta\hat{p}_{\rm out}^{(0)})^2\rangle = 1/4$; (ii) $\langle(\Delta\hat{x}_{\rm out})^2\rangle$; (iii) $\langle(\Delta\hat{p}_{\rm out})^2\rangle$. (b) The measurement results of the correlation between the output and reference modes. (i) $\langle(\Delta\hat{x}_{\rm ref}^{(0)} - \Delta\hat{x}_{\rm out}^{(0)})^2\rangle = \langle(\Delta\hat{p}_{\rm ref}^{(0)} + \Delta\hat{p}_{\rm in}^{(0)})^2\rangle = 1/2$; (ii) $\langle(\Delta\hat{x}_{\rm ref} - \Delta\hat{x}_{\rm out})^2\rangle$; (iii) $\langle(\Delta\hat{p}_{\rm ref} + \Delta\hat{p}_{\rm out})^2\rangle$. The measurement conditions are the same as those of Fig. 6.

obtained for x and p quadratures, respectively (Fig. 8a). Similarly, the noise levels for the input mode are 5.19 ± 0.13 and 4.37 ± 0.14 dB for x and p quadratures (not shown). By making electrical subtraction or summation of the homodyne outputs, the noise levels of -3.19 ± 0.13 and -4.19 ± 0.14 dB are observed for x and p quadratures as shown in Fig. 8b. From these values, the measured variance of $\Delta_{\rm ref,in} = 0.43 \pm 0.01 < 1$ is obtained. This result shows the existence of quantum entanglement between the input and the reference.

Then we proceed to the experiment of teleportation of quantum entanglement. The measurement results for the output are shown in Fig. 9. Note that the state in the reference mode does not change in the process. For the output mode, the noise levels of 6.06 ± 0.12 and 5.47 ± 0.14 dB are obtained for x and p quadratures, respectively (Fig. 9a). Because of the imperfect teleportation process, some noises are added to the teleported state, resulting in the larger variances than that of the reference. Fig. 9b shows the results of the correlation measurement. The observed noise levels are -0.25 ± 0.13 and -0.60 ± 0.13 dB for x and p quadratures, respectively, yielding $\Delta_{\rm ref,out} = 0.91 \pm 0.02 < 1$. The result clearly shows the existence of quantum entanglement between the output and the reference, and also it is a proof of success of the teleportation of quantum entanglement.

3. Multipartite quantum entanglement and its application to multiparty quantum teleportation

3.1. Sufficient conditions for the existence of multipartite quantum entanglement

As a natural extension of Eq. (5) to tripartite quantum entanglement, following inequalities are proposed as sufficient conditions for the existence of tripartite entanglement [26], where at least two inequalities are satisfied for the existence of tripartite entanglement.

$$\begin{aligned}
&\text{I. } \langle[\Delta(\hat{x}_{\rm A} - \hat{x}_{\rm B})]^2\rangle + \langle[\Delta(\hat{p}_{\rm A} + \hat{p}_{\rm B} + g_{\rm C}\hat{p}_{\rm C})]^2\rangle < 1,\\
&\text{II. } \langle[\Delta(\hat{x}_{\rm B} - \hat{x}_{\rm C})]^2\rangle + \langle[\Delta(g_{\rm A}\hat{p}_{\rm A} + \hat{p}_{\rm B} + \hat{p}_{\rm C})]^2\rangle < 1,\\
&\text{III. } \langle[\Delta(\hat{x}_{\rm C} - \hat{x}_{\rm A})]^2\rangle + \langle[\Delta(\hat{p}_{\rm A} + g_{\rm B}\hat{p}_{\rm B} + \hat{p}_{\rm C})]^2\rangle < 1,
\end{aligned} \tag{13}$$

where $\hat{x}_i$ and $\hat{p}_i$ $(i = {\rm A}, {\rm B}, {\rm C})$ are quadrature-phase operators for modes A, B, and C, and g_i $(i = {\rm A}, {\rm B}, {\rm C})$ is an arbitrary real number.

There are two extreme cases of Eqs. (13); one is the case for $g_A = g_B = g_C = 0$, and the other is the case for $g_A = g_B = g_C = 1$. In the case of $g_A = g_B = g_C = 0$, Eqs. (13) become

$$\begin{aligned}
&\text{I. } \langle[\Delta(\hat{x}_A - \hat{x}_B)]^2\rangle + \langle[\Delta(\hat{p}_A + \hat{p}_B)]^2\rangle < 1,\\
&\text{II. } \langle[\Delta(\hat{x}_B - \hat{x}_C)]^2\rangle + \langle[\Delta(\hat{p}_B + \hat{p}_C)]^2\rangle < 1,\\
&\text{III. } \langle[\Delta(\hat{x}_C - \hat{x}_A)]^2\rangle + \langle[\Delta(\hat{p}_C + \hat{p}_A)]^2\rangle < 1.
\end{aligned} \tag{14}$$

Since these inequalities are the same as Eq. (5) which is a sufficient condition for bipartite quantum entanglement, the sufficient condition for tripartite quantum entanglement in Eqs. (14) corresponds to the existence of at least two combinations of bipartite quantum entanglement. So in the case for $g_A = g_B = g_C = 0$, there exist both bipartite and tripartite quantum entanglement like the W state of qubit cases [27]. This property is used in quantum telecloning as mentioned later.

On the other hand, in the case of $g_A = g_B = g_C = 1$, Eqs. (13) become

$$\begin{aligned}
&\text{I. } \langle[\Delta(\hat{x}_A - \hat{x}_B)]^2\rangle + \langle[\Delta(\hat{p}_A + \hat{p}_B + \hat{p}_C)]^2\rangle < 1,\\
&\text{II. } \langle[\Delta(\hat{x}_B - \hat{x}_C)]^2\rangle + \langle[\Delta(\hat{p}_A + \hat{p}_B + \hat{p}_C)]^2\rangle < 1,\\
&\text{III. } \langle[\Delta(\hat{x}_C - \hat{x}_A)]^2\rangle + \langle[\Delta(\hat{p}_A + \hat{p}_B + \hat{p}_C)]^2\rangle < 1.
\end{aligned} \tag{15}$$

In the case of the left-hand side of Eqs. (15) approaches to zero, the overall state approaches CV version of Greenberger–Horn–Zeilinger (GHZ) state [28], which is a simultaneous eigenstate of four commuting operators $\hat{x}_A - \hat{x}_B$, $\hat{x}_B - \hat{x}_C$, $\hat{x}_C - \hat{x}_A$, and $\hat{p}_A + \hat{p}_B + \hat{p}_C$. Here the eigenvalues are $x_A - x_B = 0$, $x_B - x_C = 0$, $x_C - x_A = 0$, and $p_A + p_B + p_C = 0$ and then the CV GHZ state is

$$|\mathrm{GHZ}\rangle = c\int_{-\infty}^{\infty} \mathrm{d}x\, |x\rangle_A \otimes |x\rangle_B \otimes |x\rangle_C. \tag{16}$$

When the overall state is close to the CV GHZ state, no bipartite quantum entanglement exists and it is a *genuine* tripartitely entangled state. In a quantum teleportation network mentioned later, this property is used intentionally.

3.2. Creation of the GHZ type CV tripartite quantum entanglement and realization of a quantum teleportation network

Fig. 10 shows an experimental setup for creation of the CV GHZ type state and testing Eqs. (15) [26,29,30]. The created state is represented with quadrature-phase operators as follows [30]:

$$\begin{aligned}
\hat{x}_A &= \frac{1}{\sqrt{3}} e^{+r_1} \hat{x}_1^{(0)} + \sqrt{\frac{2}{3}} e^{-r_2} \hat{x}_2^{(0)},\\
\hat{p}_A &= \frac{1}{\sqrt{3}} e^{-r_1} \hat{p}_1^{(0)} + \sqrt{\frac{2}{3}} e^{+r_2} \hat{p}_2^{(0)},\\
\hat{x}_B &= \frac{1}{\sqrt{3}} e^{+r_1} \hat{x}_1^{(0)} - \frac{1}{\sqrt{6}} e^{-r_2} \hat{x}_2^{(0)} + \frac{1}{\sqrt{2}} e^{-r_3} \hat{x}_3^{(0)},\\
\hat{p}_B &= \frac{1}{\sqrt{3}} e^{-r_1} \hat{p}_1^{(0)} - \frac{1}{\sqrt{6}} e^{+r_2} \hat{p}_2^{(0)} + \frac{1}{\sqrt{2}} e^{+r_3} \hat{p}_3^{(0)},\\
\hat{x}_C &= \frac{1}{\sqrt{3}} e^{+r_1} \hat{x}_1^{(0)} - \frac{1}{\sqrt{6}} e^{-r_2} \hat{x}_2^{(0)} - \frac{1}{\sqrt{2}} e^{-r_3} \hat{x}_3^{(0)},\\
\hat{p}_C &= \frac{1}{\sqrt{3}} e^{-r_1} \hat{p}_1^{(0)} - \frac{1}{\sqrt{6}} e^{+r_2} \hat{p}_2^{(0)} - \frac{1}{\sqrt{2}} e^{+r_3} \hat{p}_3^{(0)}.
\end{aligned} \tag{17}$$

where superscript (0) denotes a vacuum, r_1, r_2, and r_3 are squeezing parameters for the respective modes. When r_1, r_2, and r_3 go to infinity, the state approaches to the CV GHZ state.

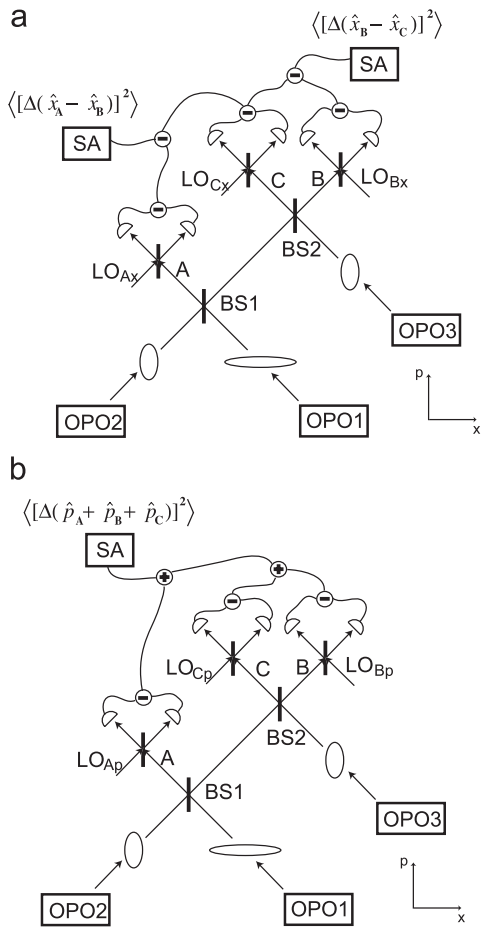


Fig. 10. Experimental setup for creation of the CV GHZ type state and testing Eq. (15) [26,29,30]. All beam splitters except for BS1 are half beam splitters. BS1 is a 1:2 beam splitter.

Fig. 11 shows examples of the results for the experiments of Fig. 10 reported by Aoki et al. [29]. From these results, we can calculate the left-hand sides of Eqs. (15) and get [29]

$$\begin{aligned}
&\text{I. } \langle[\Delta(\hat{x}_A - \hat{x}_B)]^2\rangle + \langle[\Delta(\hat{p}_A + \hat{p}_B + \hat{p}_C)]^2\rangle = 0.851 \pm 0.062 < 1,\\
&\text{II. } \langle[\Delta(\hat{x}_B - \hat{x}_C)]^2\rangle + \langle[\Delta(\hat{p}_A + \hat{p}_B + \hat{p}_C)]^2\rangle = 0.840 \pm 0.065 < 1,\\
&\text{III. } \langle[\Delta(\hat{x}_C - \hat{x}_A)]^2\rangle + \langle[\Delta(\hat{p}_A + \hat{p}_B + \hat{p}_C)]^2\rangle = 0.867 \pm 0.062 < 1.
\end{aligned} \tag{18}$$

From the calculations, we can see the satisfaction of Eqs. (15) and the existence of tripartite entanglement.

A quantum teleportation network proposed by van Loock and Braunstein is an extended version of Braunstein–Kimble scheme of CV quantum teleportation [30]. The essential difference is the requirement for quantum entanglement. For Braunstein–Kimble scheme we need to have bipartite quantum entanglement between a sender and a receiver. On the other hand, we need to have the CV GHZ type state for the quantum teleportation network, which has tripartite entanglement among all three members, but does not have bipartite entanglement between any combinations in the network. In the network, a general quantum state is teleported between two members with the help of the third one as shown in Fig. 12 [30]. Here three members Alice, Bob, and Claire share the CV GHZ type state. In the case of teleportation from Alice to Bob, for example, teleportation does not succeed without the information from Claire, because there is no bipartite quantum entanglement between Alice and Bob. The information from Claire must be needed for success of teleportation from Alice to Bob in the network.

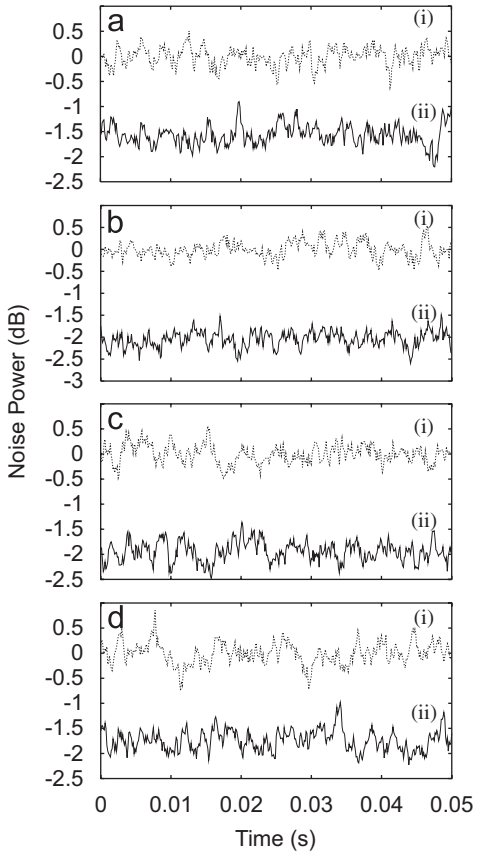


Fig. 11. Examples of the results for the experiments shown in Fig. 10 [29]. (a) $\langle[\Delta(\hat{x}_A - \hat{x}_B)]^2\rangle$, (b) $\langle[\Delta(\hat{x}_B - \hat{x}_C)]^2\rangle$, (c) $\langle[\Delta(\hat{x}_C - \hat{x}_A)]^2\rangle$, (d) $\langle[\Delta(\hat{p}_A + \hat{p}_B + \hat{p}_C)]^2\rangle$. (i) no entanglement (all inputs are vacua), (ii) some entanglement (all inputs are squeezed vacua). $\langle[\Delta(\hat{x}_1^{(0)} - \hat{x}_2^{(0)})]^2\rangle = \langle[\Delta(\hat{x}_2^{(0)} - \hat{x}_3^{(0)})]^2\rangle = \langle[\Delta(\hat{x}_3^{(0)} - \hat{x}_1^{(0)})]^2\rangle = 1/2$ and $\langle[\Delta(\hat{p}_1^{(0)} + \hat{p}_2^{(0)} + \hat{p}_3^{(0)})]^2\rangle = 3/4$.

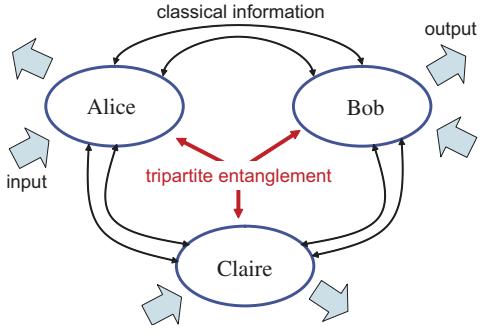


Fig. 12. A quantum teleportation network. Teleportation can be performed between any combinations in the network with the classical information from the third member.

Here we present the experiment of a quantum teleportation network performed by Yonezawa et al. [31]. Fig. 13 shows the scheme of the experimental setup. The CV GHZ type state is distributed to Alice, Bob, and Claire. Here we consider the case of teleportation from Alice to Bob with the help of Claire.

As the same as Braunstein–Kimble scheme of CV quantum teleportation, Alice measures $\hat{x}_- = (\hat{x}_{in} - \hat{x}_A)/\sqrt{2}$ and $\hat{p}_+ = (\hat{p}_{in} + \hat{p}_A)/\sqrt{2}$ as shown in Fig. 13, where an input mode is represented by $(\hat{x}_{in}, \hat{p}_{in})$ and Alice's entangled

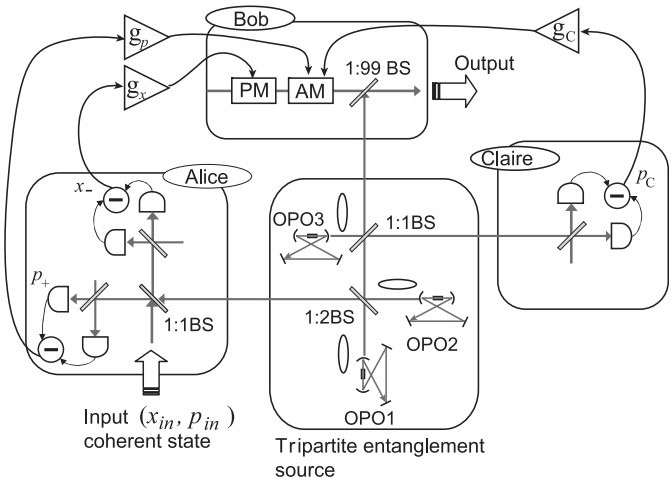


Fig. 13. Experimental setup for a quantum teleportation network. This schematic indicates teleportation from Alice to Bob with the help of Claire [31]. The information gains g_x and g_p are set to be one.

mode is represented by $(\hat{x}_A, \hat{p}_A)$. Claire measures her p quadrature $\hat{p}_C$. Bob's mode B before these measurements is $\hat{x}_B = \hat{x}_{in} - (\hat{x}_A - \hat{x}_B) - \sqrt{2}\hat{x}_-$ and $\hat{p}_B = \hat{p}_{in} + (\hat{p}_A + \hat{p}_B + g_C\hat{p}_C) - \sqrt{2}\hat{p}_+ - g_C\hat{p}_C$, where g_C is a information gain of the classical channel from Claire to Bob. After the Alice's and Claire's measurements, operators $\hat{x}_-$, $\hat{p}_+$, and $\hat{p}_C$ collapse and reduce to certain values. Receiving their measurement results through classical channels, Bob displace his mode as $\hat{x}_B \to \hat{x}_{out} = \hat{x}_B + \sqrt{2}g_x x_-$ and $\hat{p}_B \to \hat{p}_{out} = \hat{p}_B + \sqrt{2}g_p p_+ + g_C p_C$. Here g_x and g_p are information gains of classical channels from Alice to Bob and they are usually $g_x = g_p = 1$. Thus with Eq. (17) the teleported output mode can be calculated as [30]

$$\begin{aligned}
\hat{x}_{out} &= \hat{x}_{in} - (\hat{x}_A - \hat{x}_B) \\
&= \hat{x}_{in} - \sqrt{\frac{2}{3}}e^{-r_2}\hat{x}_2^{(0)} + \frac{1}{\sqrt{2}}e^{-r_3}\hat{x}_3^{(0)}, \\
\hat{p}_{out} &= \hat{p}_{in} + (\hat{p}_A + \hat{p}_B + g_C\hat{p}_C) \\
&= \hat{p}_{in} + \frac{2+g_C}{\sqrt{3}}e^{-r_1}\hat{p}_1^{(0)} + \frac{1-g_C}{\sqrt{6}}e^{r_2}\hat{p}_2^{(0)} + \frac{1-g_C}{\sqrt{2}}e^{r_3}\hat{p}_3^{(0)}.
\end{aligned} \tag{19}$$

For a coherent state input and in the case of $r_1 = r_2 = r_3 = r$, the teleportation fidelity F_{coh} is calculated as [30]

$$\begin{aligned}
F_{coh} &= \frac{2}{\sqrt{(1+4\langle(\Delta\hat{x}_{out})^2\rangle)(1+4\langle(\Delta\hat{p}_{out})^2\rangle)}} \\
&= \frac{1}{\sqrt{1+e^{-2r}}\sqrt{1+\frac{(2+g_C)^2}{6}e^{-2r}+\frac{(1-g_C)^2}{3}e^{2r}}}.
\end{aligned} \tag{20}$$

In the ideal case ($r \to \infty$), the teleportation is perfect ($F_{coh} \to 1$). In other cases, the fidelity F_{coh} is a function of the squeezing parameter (r) and the information gain of classical channel from Claire to Bob (g_C). Again, the success criterion is $F_{coh} > 1/2$.

Fig. 14 shows experimental results for quantum teleportation from Alice to Bob with the help of Claire [31]. The information gain of classical channel from Claire to Bob is $g_C = 1.02 \pm 0.03$. The noise level of p quadrature of the teleported state for a vacuum input is 3.3 dB higher than the corresponding vacuum noise level as shown in the figure, while that of x quadrature is 3.5 dB (not shown). With these values and Eq. (20), we can get the fidelity of 0.63 beyond the classical limit of 1/2. The fact clearly shows the success of teleportation from Alice to Bob with the help of Claire.

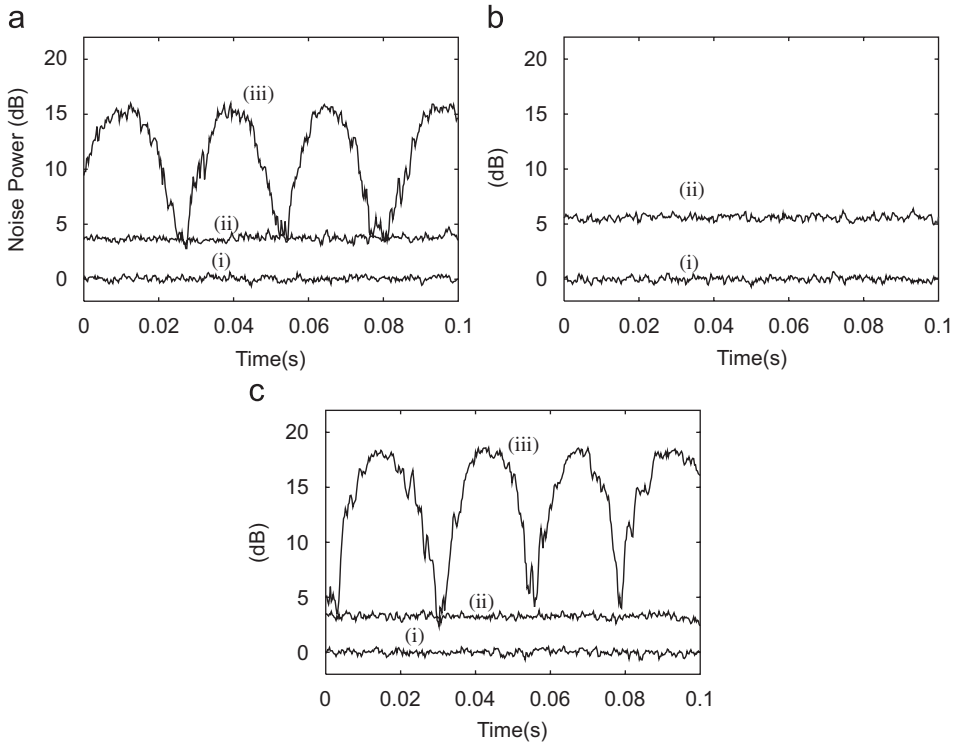


Fig. 14. Experimental results for quantum teleportation from Alice to Bob with the help of Claire [31]. (a) Alice's measurement results for p quadrature. (i) $\langle(\Delta\hat{p}_+^{(0)})^2\rangle = 1/4$ where (0) denotes that every initial state is a vacuum, (ii) $\langle(\hat{p}_+)^2\rangle$ $(=\langle(\Delta\hat{p}_+)^2\rangle)$ for a vacuum input, (iii) $\langle(\hat{p}_+)^2\rangle$ for a coherent state input with the phase scanned. (b) Claire's measurement results for p quadrature. (i) $\langle(\Delta\hat{p}_C^{(0)})^2\rangle = 1/4$, (ii) $\langle(\hat{p}_C)^2\rangle$ $(=\langle(\Delta\hat{p}_C)^2\rangle)$ for a vacuum input, (c) Measurement results of the teleported state for p quadrature. (i) $\langle(\Delta\hat{p}_{out}^{(0)})^2\rangle = 1/4$, (ii) $\langle(\hat{p}_{out})^2\rangle(=\langle(\Delta\hat{p}_{out})^2\rangle)$ for a vacuum input, (iii) $\langle(\hat{p}_{out})^2\rangle$ for the coherent state input. The measurement conditions are the same as those of Fig. 6.

The information gain dependences of the fidelity are plotted with the theoretical curves in Fig. 15. Quantum teleportation is performed in three combinations of a sender and a receiver. There are two important points in the figure. One is, of course, quantum teleportation succeeds in three combinations of a sender and a receiver with the quantum entanglement. In other words, it is a proof of the existence of the tripartite quantum entanglement in the system. The other important point is that the quantum teleportation fails without the information from the third member, which corresponds to the case of $g_i = 0$ $(i = \mathrm{A}, \mathrm{B}, \mathrm{C})$. This fact shows that there is no bipartite quantum entanglement and the shared entangled state is CV GHZ type.

3.3. Quantum telecloning

Quantum telecloning is an application of another type of tripartite quantum entanglement; the CV W-like state, which has bipartite and tripartite quantum entanglement simultaneously.

Quantum telecloning [32] is a quantum information protocol combining cloning and teleporting into a single new primitive. Here the cloning process produces approximate clones beyond the classical limit of fidelity but they never reach the perfect ones because of the no-cloning theorem [51]. The protocol offers significant technical advantages over the two-step local cloning plus teleportation strategy. In particular, in the case of coherent state telecloning, only finite entanglement is required for generating remote clones with optimal fidelity [33].

In fact, quantum telecloning generalizes quantum teleportation with multiple receivers [32]. In quantum teleportation, bipartite quantum entanglement shared by two parties (Alice and Bob) enables them to teleport an unknown quantum state from Alice to Bob by communicating only through classical channels as shown in previous subsections. If three parties (Alice, Bob, and Claire) share an appropriate tripartite entangled state, Alice is able to teleport an unknown

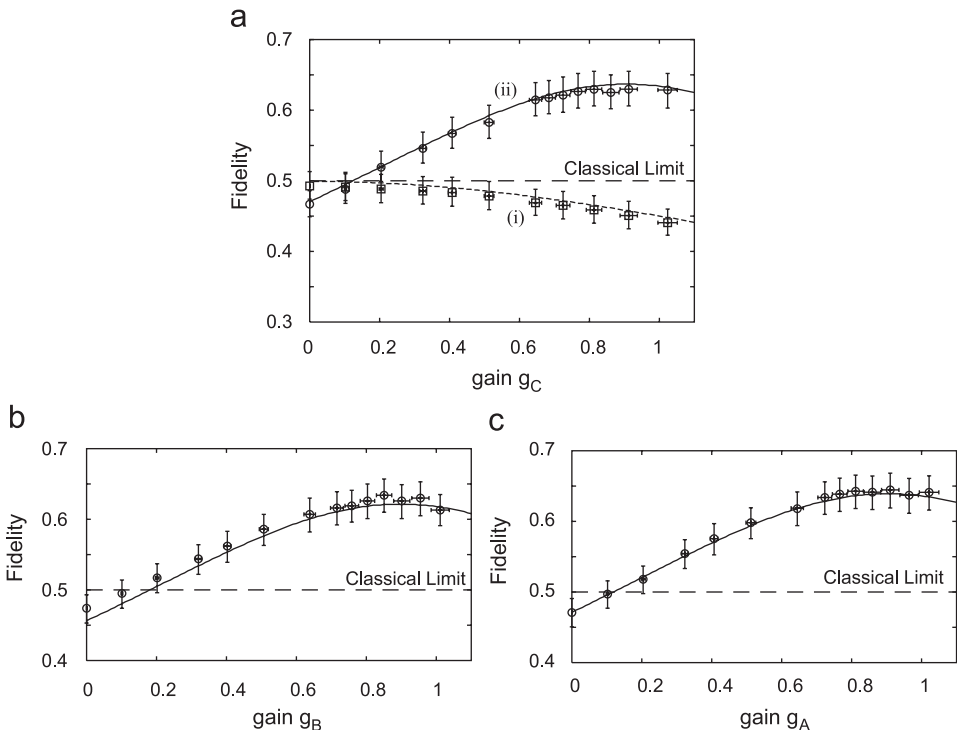


Fig. 15. Information gain dependences of the fidelities [31]. (a) The fidelity of teleportation from Alice to Bob with the help of Claire. (i) The teleportation without quantum entanglement ($r = 0$). (ii) The teleportation with the CV GHZ type tripartite quantum entanglement ($r > 0$). The best fidelity is obtained at $g_C \simeq 0.9$. The solid curves represent the theoretical ones calculated with experimental parameters and Eq. (20). (b) The fidelity of teleportation from Alice to Claire with the help of Bob. The best fidelity is obtained at $g_B \simeq 0.9$. (b) The fidelity of teleportation from Claire to Bob with the help of Alice. The best fidelity is obtained at $g_A \simeq 0.9$.

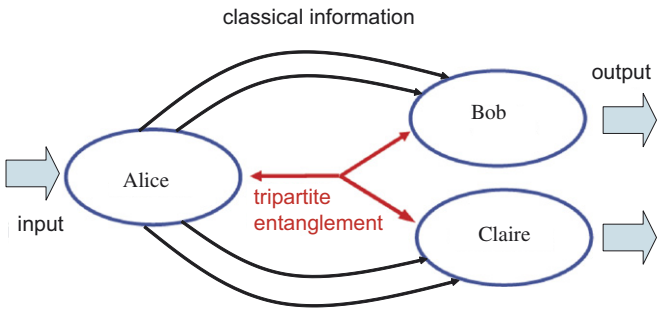


Fig. 16. $1 \rightarrow 2$ quantum telecloning. Alice is a sender and Bob and Claire are two receivers.

quantum state to Bob and Claire simultaneously. This is called '$1 \rightarrow 2$ quantum telecloning' (Fig. 16). More generally, quantum telecloning to an arbitrary number of receivers ($1 \rightarrow n$ quantum telecloning) can be performed by using multipartite quantum entanglement.

The heart of quantum telecloning is the multipartite quantum entanglement shared among the sender and receivers. Without multipartite quantum entanglement, only the corresponding two-step protocol is possible: first the sender makes clones locally [34], and then sends them to each receiver with bipartite quantum teleportation [10,20] (or vice versa, teleporting followed by local cloning). The two-step protocol would require maximal bipartite quantum entanglement for optimal fidelity teleportation (which for CV quantum teleportation requires infinite amount of squeezing). Surprisingly,

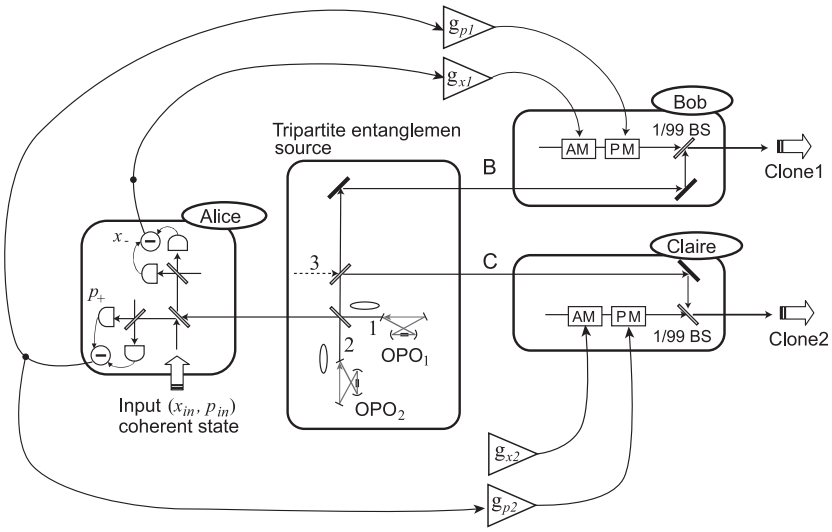


Fig. 17. The experimental setup for quantum telecloning from Alice to Bob and Claire to produce Clone 1 and 2. Modes i and ii are squeezed vacua and mode iii is a vacuum. The g's are normalized information gains of classical channels. AM denotes an amplitude modulator and PM denotes a phase modulator. A 99%-reflectance beam splitter (denoted 1/99 BS) enables displacement transformations in phase space [20].

CV quantum telecloning of coherent states requires only finite squeezing to achieve the same optimal fidelity [33]. In fact, the level of squeezing needed for optimal telecloning of coherent states is the reach of current technology [24,25].

Here we present the experiment of $1 \to 2$ quantum telecloning of coherent states performed by Koike et al. [35]. Fig. 17 shows the experimental setup for quantum telecloning from Alice to Bob and Claire to produce clones 1 and 2.

The scheme for creating the tripartite entanglement for quantum telecloning [33] is shown in the center of Fig. 17. Two squeezed vacua are first combined with a half beam splitter and then one of the output beams is divided into two beams (B,C) with another half beam splitter. The three output modes $(\hat{x}_j, \hat{p}_j)$ $(j = \mathrm{A}, \mathrm{B}, \mathrm{C})$ are the following:

$$
\begin{aligned}
\hat{x}_{\mathrm{A}} &= \frac{1}{\sqrt{2}} e^{+r_1} \hat{x}_1^{(0)} + \frac{1}{\sqrt{2}} e^{-r_2} \hat{x}_2^{(0)}, \\
\hat{p}_{\mathrm{A}} &= \frac{1}{\sqrt{2}} e^{-r_1} \hat{p}_1^{(0)} + \frac{1}{\sqrt{2}} e^{+r_2} \hat{p}_2^{(0)}, \\
\hat{x}_{\mathrm{B}} &= \frac{1}{2} e^{+r_1} \hat{x}_1^{(0)} - \frac{1}{2} e^{-r_2} \hat{x}_2^{(0)} + \frac{1}{\sqrt{2}} \hat{x}_3^{(0)}, \\
\hat{p}_{\mathrm{B}} &= \frac{1}{2} e^{-r_1} \hat{p}_1^{(0)} - \frac{1}{2} e^{+r_2} \hat{p}_2^{(0)} + \frac{1}{\sqrt{2}} \hat{p}_3^{(0)}, \\
\hat{x}_{\mathrm{C}} &= \frac{1}{2} e^{+r_1} \hat{x}_1^{(0)} - \frac{1}{2} e^{-r_2} \hat{x}_2^{(0)} - \frac{1}{\sqrt{2}} \hat{x}_3^{(0)}, \\
\hat{p}_{\mathrm{C}} &= \frac{1}{2} e^{-r_1} \hat{p}_1^{(0)} - \frac{1}{2} e^{+r_2} \hat{p}_2^{(0)} - \frac{1}{\sqrt{2}} \hat{p}_3^{(0)}.
\end{aligned}
\tag{21}
$$

Here, modes A and B and modes A and C are bipartitely entangled and modes A, B, and C are tripartitely entangled. This can be verified by applying the sufficient inseparability criteria for a bipartite case Eq. (5) and a tripartite case (Eqs. (14)). In the present situation with $r_1 = r_2 = r$, the criteria are:

$$
\langle [\Delta(\hat{x}_{\mathrm{A}} - \hat{x}_{\mathrm{B}})]^2 \rangle + \langle [\Delta(\hat{p}_{\mathrm{A}} + \hat{p}_{\mathrm{B}})]^2 \rangle = \left(\frac{1 - \sqrt{2}}{2} \right)^2 \frac{e^{2r}}{2} + \left(\frac{1 + \sqrt{2}}{2} \right)^2 \frac{e^{-2r}}{2} + \frac{1}{4} < 1,
$$

$$\langle[\Delta(\hat{x}_{\rm A}-\hat{x}_{\rm C})]^2\rangle+\langle[\Delta(\hat{p}_{\rm A}+\hat{p}_{\rm C})]^2\rangle=\left(\frac{1-\sqrt{2}}{2}\right)^2\frac{e^{2r}}{2}+\left(\frac{1+\sqrt{2}}{2}\right)^2\frac{e^{-2r}}{2}+\frac{1}{4}<1. \tag{22}$$

The left-hand side of the inequality can be minimized when $e^{-2r}=(\sqrt{2}-1)/(\sqrt{2}+1)$ (-7.7 dB squeezing). By using these tripartitely entangled modes, sender Alice can perform quantum telecloning of a coherent state input to two receivers Bob and Claire to produce clone 1 and 2 at their sites. In other words, the success of quantum telecloning is a sufficient condition for the existence of this type of quantum entanglement.

The scheme of the quantum telecloning is almost the same as the Braunstein–Kimble scheme of CV quantum teleportation. Alice measures $\hat{x}_-=(\hat{x}_{\rm in}-\hat{x}_{\rm A})/\sqrt{2}$ and $\hat{p}_+=(\hat{p}_{\rm in}+\hat{p}_{\rm A})/\sqrt{2}$ as shown in Fig. 17. Bob's mode B and Claire's mode C before the measurement are:

$$\begin{aligned}
\hat{x}_{\rm B}&=\hat{x}_{\rm in}-(\hat{x}_{\rm A}-\hat{x}_{\rm B})-\sqrt{2}\hat{x}_-,\\
\hat{p}_{\rm B}&=\hat{p}_{\rm in}+(\hat{p}_{\rm A}+\hat{p}_{\rm B})-\sqrt{2}\hat{p}_+,\\
\hat{x}_{\rm C}&=\hat{x}_{\rm in}-(\hat{x}_{\rm A}-\hat{x}_{\rm C})-\sqrt{2}\hat{x}_-,\\
\hat{p}_{\rm C}&=\hat{p}_{\rm in}+(\hat{p}_{\rm A}+\hat{p}_{\rm C})-\sqrt{2}\hat{p}_+.
\end{aligned} \tag{23}$$

After the Alice's measurement, operators $\hat{x}_-$ and $\hat{p}_+$ collapse and reduce to certain values. Receiving her measurement results through classical channels, Bob and Claire displace his and her modes as $\hat{x}_{\rm B}\to\hat{x}_{\rm clone1}=\hat{x}_{\rm B}+\sqrt{2}g_x x_-$, $\hat{p}_{\rm B}\to\hat{p}_{\rm clone1}=\hat{p}_{\rm B}+\sqrt{2}g_p p_+$, $\hat{x}_{\rm C}\to\hat{x}_{\rm clone2}=\hat{x}_{\rm C}+\sqrt{2}g_x x_-$, and $\hat{p}_{\rm C}\to\hat{p}_{\rm clone2}=\hat{p}_{\rm C}+\sqrt{2}g_p p_+$. Here g_x and g_p are information gains of classical channels from Alice to Bob and Claire, and they are usually $g_x=g_p=1$. Thus with Eq.(21) the telecloned output modes become [33]

$$\begin{aligned}
\hat{x}_{\rm clone1}&=\hat{x}_{\rm in}-(\hat{x}_{\rm A}-\hat{x}_{\rm B})\\
&=\hat{x}_{\rm in}+\frac{1-\sqrt{2}}{2}e^r\hat{x}_1^{(0)}-\frac{1+\sqrt{2}}{2}e^{-r}\hat{x}_2^{(0)}+\frac{1}{\sqrt{2}}\hat{x}_3^{(0)},\\
\hat{p}_{\rm clone1}&=\hat{p}_{\rm in}+(\hat{p}_{\rm A}+\hat{p}_{\rm B})\\
&=\hat{p}_{\rm in}+\frac{1+\sqrt{2}}{2}e^{-r}\hat{p}_1^{(0)}-\frac{1-\sqrt{2}}{2}e^r\hat{p}_2^{(0)}+\frac{1}{\sqrt{2}}\hat{p}_3^{(0)},\\
\hat{x}_{\rm clone2}&=\hat{x}_{\rm in}-(\hat{x}_{\rm A}-\hat{x}_{\rm C})\\
&=\hat{x}_{\rm in}+\frac{1-\sqrt{2}}{2}e^r\hat{x}_1^{(0)}-\frac{1+\sqrt{2}}{2}e^{-r}\hat{x}_2^{(0)}-\frac{1}{\sqrt{2}}\hat{x}_3^{(0)},\\
\hat{p}_{\rm clone2}&=\hat{p}_{\rm in}+(\hat{p}_{\rm A}+\hat{p}_{\rm C})\\
&=\hat{p}_{\rm in}+\frac{1+\sqrt{2}}{2}e^{-r}\hat{p}_1^{(0)}-\frac{1-\sqrt{2}}{2}e^r\hat{p}_2^{(0)}-\frac{1}{\sqrt{2}}\hat{p}_3^{(0)},
\end{aligned} \tag{24}$$

where subscript 3 denotes a vacuum input to the second beam splitter in the tripartite entanglement source in Fig. 17.

For a coherent state input, the telecloning fidelity $F_{\rm coh}$ is again calculated as:

$$\begin{aligned}
F_{\rm coh}&=\frac{2}{\sqrt{(1+4\langle(\Delta\hat{x}_{\rm clone1,2})^2\rangle)(1+4\langle(\Delta\hat{p}_{\rm clone1,2})^2\rangle)}}\\
&=\frac{1}{\left(\frac{1-\sqrt{2}}{2}\right)^2\frac{e^{2r}}{2}+\left(\frac{1+\sqrt{2}}{2}\right)^2\frac{e^{-2r}}{2}+\frac{5}{4}}.
\end{aligned} \tag{25}$$

The fidelity can be optimized and $F_{\rm coh}=2/3$ when $e^{-2r}=(\sqrt{2}-1)/(\sqrt{2}+1)$ (-7.7 dB squeezing). The important thing here is that we can get the optimum fidelity of 2/3 with finite squeezing of -7.7 dB which is achievable with

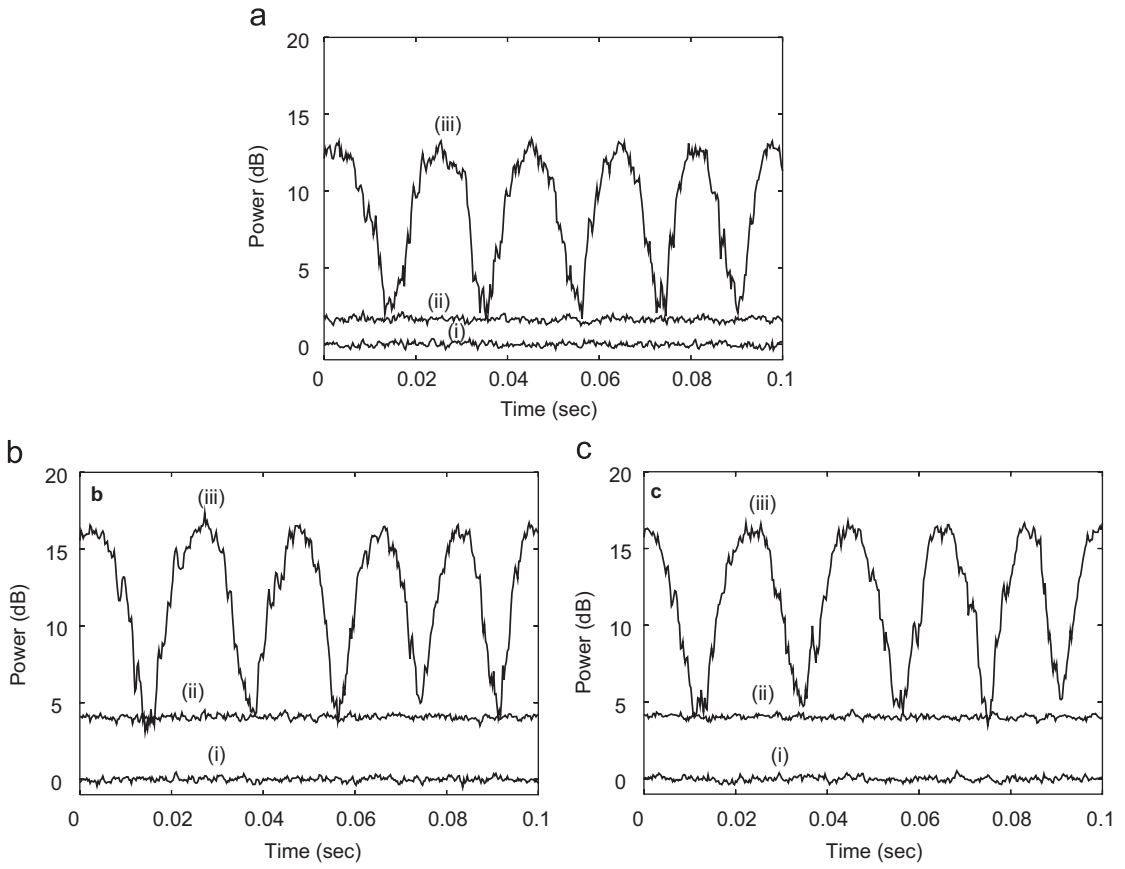


Fig. 18. Experimental results for quantum telecloning from Alice to Bob and Claire to produce Clone 1 and 2 [35]. All traces are normalized to the corresponding vacuum noise levels. (a) Alice's measurement results for p quadrature. Trace i, the corresponding vacuum noise level $\langle(\Delta\hat{p}^{(0)})^2\rangle = 1/4$. Trace ii, the measurement result of a vacuum input $\langle(\hat{p}'_+)^2\rangle$ where $\hat{p}'_+ = (\hat{p}^{(0)}_{\rm in} + \hat{p}_{\rm A})/\sqrt{2}$. Trace iii, the measurement result of a coherent state input $\langle(\hat{p}_+)^2\rangle$ with the phase scanned. (b),(c) The measurement results of the telecloned states at Bob (b) and Claire (c) for p quadratures (x quadratures are not shown). Trace i, the corresponding vacuum noise levels. Trace ii, the telecloned states for a vacuum input $\langle(\Delta\hat{p}_{{\rm clone}1,2})^2\rangle$. Trace iii, the telecloned states for a coherent state input. The measurement frequency is centered at 1 MHz, and the resolution and video bandwidths are 30 kHz and 300 Hz, respectively. All traces except for trace iii are averaged 20 times.

current technology [24,25]. Note that the fidelity of 2/3 is called as "no-cloning limit" as mentioned in the previous section.

Fig. 18 shows the results of telecloning experiments [35]. Quantum telecloning is performed for two types of input states: a vacuum state and a coherent state that is created by applying electro-optic modulation to a very weak carrier beam. Alice's states are shown in Fig. 18a, and Bob and Claire's output states are shown in Fig. 18b and c. Trace ii of Fig. 18a shows Alice's p-quadrature measurement for a vacuum input, $\langle(\hat{p}'_+)^2\rangle$, where $\hat{p}'_+ = (\hat{p}^{(0)}_{\rm in} + \hat{p}_{\rm A})/\sqrt{2}$. Note that $\langle\hat{p}^{(0)}_{\rm in}\rangle = \langle\hat{p}_{\rm A}\rangle = 0$; thus $\langle(\hat{p}'_+)^2\rangle = \langle(\Delta\hat{p}'_+)^2\rangle = \langle(\Delta\hat{p}_+)^2\rangle$, because the vacuum is a zero-amplitude coherent state. The noise level is 2.1 dB higher compared to the vacuum noise level $\langle(\Delta\hat{p}^{(0)})^2\rangle = 1/4$, due to the "entangled noise" $\hat{p}_{\rm A}$. This noise is cancelled to some extent by the tripartite entanglement. Trace iii in Fig. 18a shows Alice's coherent-state input with the phase scanned. Consistent with the above discussion on the variance of a coherent state input, the troughs of trace iii are level with trace ii within experimental accuracy. Note that Alice's 50/50 beam splitter reduces the amplitude of the measured state (i.e., the peaks of trace iii) by 3 dB relative to the input state.

Figs. 18b and c show the measurement results of the telecloned states. Traces ii show the results for a vacuum input, $\langle(\Delta\hat{p}_{1,2})^2\rangle$. The noise level for clone 1 is 4.06 ± 0.17 dB and that for clone 2 is 4.03 ± 0.15 dB. The x-quadratures $\langle(\Delta\hat{x}_{1,2})^2\rangle$ are also measured and 3.74 ± 0.15 dB for clone 1 and 3.79 ± 0.15 dB for Clone 2 (not shown) are obtained.

By using these values we can calculate the fidelity from Eq. (25), and find $F = 0.58 \pm 0.01$ for both teleclones. Since this fidelity exceeds the classical limit of telecloning of $1/2$, $1 \rightarrow 2$ quantum telecloning of coherent states is successfully demonstrated. Furthermore, these results are operational evidence for the existence of the tripartite quantum entanglement which has bipartite and tripartite quantum entanglement.

4. Universal CV quantum information processing

Quantum information processing is represented by a unitary transformation as shown in Fig. 19, where $|\psi(0)\rangle$ is an input state, $|\psi(t)\rangle$ is the output state, and $\hat{U}$ represents a quantum information processor. The unitary transformation is represented by some Hamiltonian $\hat{H}$ as follows:

$$|\psi(t)\rangle = \hat{U}|\psi(0)\rangle = \mathrm{e}^{-\mathrm{i}\frac{\hat{H}}{\hbar}t}|\psi(0)\rangle. \tag{26}$$

It follows that we have to prepare arbitrary Hamiltonians for universal quantum information processing. For universal CV quantum information processing, it is known that the following five Hamiltonians are enough to prepare arbitrary Hamiltonians [36]:

(1) $\frac{1}{2i}(\hat{a}_1^\dagger \hat{a}_2 - \hat{a}_1 \hat{a}_2^\dagger)$: a beam splitter,
(2) $i(\alpha \hat{a}^\dagger - \alpha^* \hat{a})$: displacement in phase space,
(3) $\hat{a}^\dagger \hat{a}$: phase shift,
(4) $i(\hat{a}^2 - \hat{a}^{\dagger 2})$: squeezing,
(5) $\hat{a}_1^\dagger \hat{a}_1 \hat{a}_2^\dagger \hat{a}_2$: optical Kerr effect (This Hamiltonian can be replaced by any third order nonlinear Hamiltonians.)

The trick is supported by the following relation:

$$\mathrm{e}^{\mathrm{i}\hat{H}_2\delta t}\mathrm{e}^{\mathrm{i}\hat{H}_1\delta t}\mathrm{e}^{-\mathrm{i}\hat{H}_2\delta t}\mathrm{e}^{-\mathrm{i}\hat{H}_1\delta t} \sim \mathrm{e}^{[\hat{H}_1,\hat{H}_2]\delta t^2}. \tag{27}$$

From the relation, we can see that when we have two Hamiltonians $\hat{H}_1$ and $\hat{H}_2$ we can synthesize Hamiltonian $\mathrm{i}[\hat{H}_1, \hat{H}_2]$. So we can prepare arbitrary Hamiltonians from the above five Hamiltonians by using the relation [36].

The Hamiltonians (1)–(3) are easily prepared in quantum optics experiments. But preparing the Hamiltonians (4) and (5) are not easy. Squeezing operation for a vacuum state is not difficult in these days [24], but it is difficult for arbitrary input states because of inevitable losses in squeezing process. Also, optical Kerr effect in single-photon level is very weak and it is very difficult to use the effect for arbitrary input states. To overcome these difficulties, a universal squeezer with measurement and feedforward [37] and a cubic phase gate [38] are proposed. Note that the Hamiltonian (1)–(4) makes a group on Eq. (27) and the physical operations with these Hamiltonians are called *Gaussian operations*. Similarly, the other physical operations with the Hamiltonian (5) is called *non-Gaussian operations*. Also, we should stress that the most fundamental Gaussian operation is CV quantum teleportation.

A universal squeezer for arbitrary input states is proposed by Filip et al. [37] and the scheme is shown in Fig. 20. In the scheme, an ancillary squeezed vacuum and a beam splitter are used instead of a squeezer for arbitrary input states. The scheme is described in the Heisenberg picture as follows. First, we prepare an ancillary squeezed vacuum $(\hat{x}_a, \hat{p}_a) = (e^{-r}\hat{x}_a^{(0)}, e^r \hat{p}_a^{(0)})$ with a vacuum squeezer like an OPO. Then it is combined with an input state $(\hat{x}_i, \hat{p}_i)$ at a beam splitter whose transmission coefficient is T. The output quadratures $(\hat{x}_i', \hat{p}_i')$ and $(\hat{x}_a', \hat{p}_a')$ are written as

$$\begin{aligned}
\hat{x}_i' &= \sqrt{T}\hat{x}_i + \sqrt{1-T}\hat{x}_a, \\
\hat{p}_i' &= \sqrt{T}\hat{p}_i + \sqrt{1-T}\hat{p}_a, \\
\hat{x}_a' &= \sqrt{T}\hat{x}_a - \sqrt{1-T}\hat{x}_i, \\
\hat{p}_a' &= \sqrt{T}\hat{p}_a - \sqrt{1-T}\hat{p}_i.
\end{aligned} \tag{28}$$

We measure the p quadrature of the ancillary mode and perform the feedforward, $\hat{p}_i' \rightarrow \hat{p}_i' + g\hat{p}_a'$ with gain $g = -\sqrt{(1-T)/T}$, so that the residual anti-squeezed noise of the ancillary squeezed vacuum is completely cancelled.

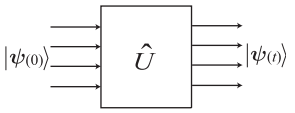


Fig. 19. Quantum information processing. $|\psi(0)\rangle$ is an input state, $|\psi(t)\rangle$ is the output state, and $\hat{U}$ represents a quantum information processor.

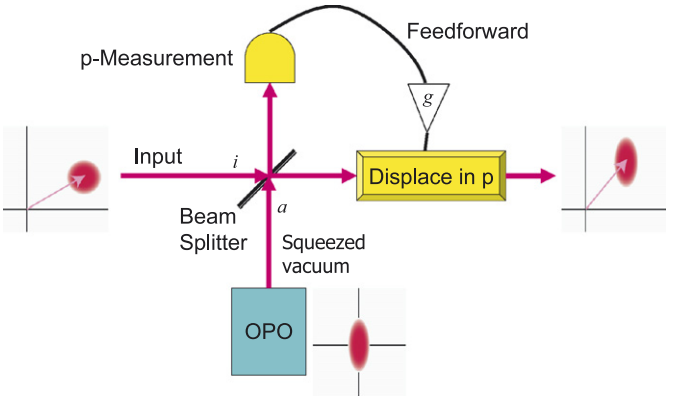


Fig. 20. A universal squeezer with measurement and feedforward [37].

Finally we get the following output quadratures:

$$\hat{x}_i' = \sqrt{T}\hat{x}_i + \sqrt{1-T}\hat{x}_a,$$
$$\hat{p}_i' = \frac{1}{\sqrt{T}}\hat{p}_i. \tag{29}$$

This resulting state is squeezed with respect to the input state, with some additional noise in the squeezed quadrature. This additional noise comes from the ancillary squeezed vacuum. In the case of $r \to \infty$ for the ancillary squeezed vacuum, the squeezing transformation becomes ideal. Moreover the squeezing degree is controllable through the transmission coefficient T. Note that we can never make noise suppression greater than that of the original ancillary squeezed vacuum. However, here our purpose is not to obtain a high level squeezing of one quadrature, but to improve the fidelity of the transformation in order to achieve tolerant quantum information processing. Realization of the universal squeezer is reported by Yoshikawa et al. for coherent state inputs [39].

A cubic phase gate [38] can replace the fifth Hamiltonian, optical Kerr effect. The action of the gate is following:

$$|\psi_{\text{out}}\rangle = \exp(\mathrm{i}\gamma\hat{x}^3)|\psi_{\text{in}}\rangle. \tag{30}$$

Realization scheme of the cubic phase gate is shown in Fig. 21. This scheme consists of four parts of Gaussian operations and photon number measurement. The first part consists of creation of EPR beams with two squeezed vacua and a beam splitter and displacement operation of one of EPR beams. This part can be regarded as the process of CV quantum teleportation, because the part includes the essential ingredients for CV quantum teleportation. The second part is a squeezer which can be realized with the universal squeezer explained above. In some sense, the universal squeezer can also be regarded as the process of CV quantum teleportation, because, in the process, quantum entanglement is created between the outputs from the beam splitter and displacement operation is performed according to the measurement result for one of the entangled beams.

The third part of the phase cubic gate is quantum non-demolition (QND) measurement which can be realized by using two universal squeezers [40–42] as shown in Fig. 22. From the same reason as the universal squeezer, this QND scheme can be regarded as the process of CV quantum teleportation. Similarly, the fourth part is a Gaussian operation which can be realized by squeezers and the linear optical elements and can be regarded as the process of CV quantum teleportation.

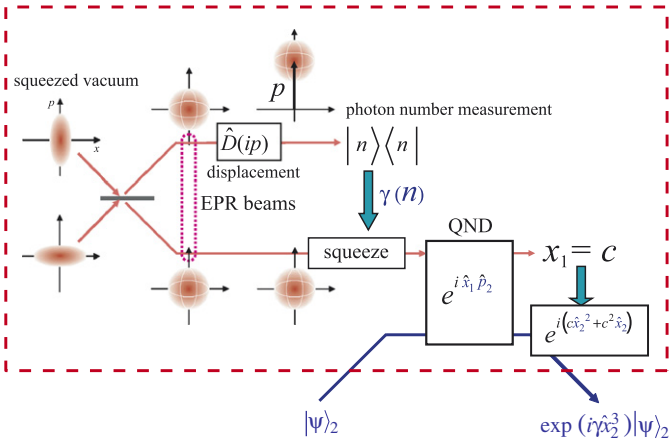


Fig. 21. Realization scheme of a cubic phase gate [38]. QND: quantum non-demolition measurement. $\gamma(n)$ is a squeezing parameter determined by the result of photon number measurement.

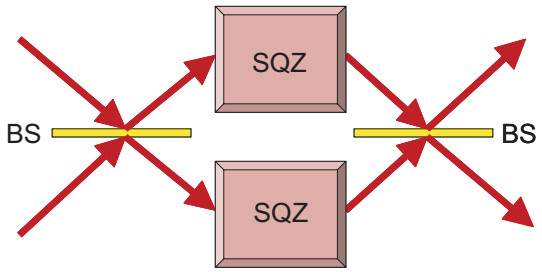


Fig. 22. Realization scheme of quantum nondemolition measurement [40–42]. BS: a beam splitter, SQZ: a squeezer.

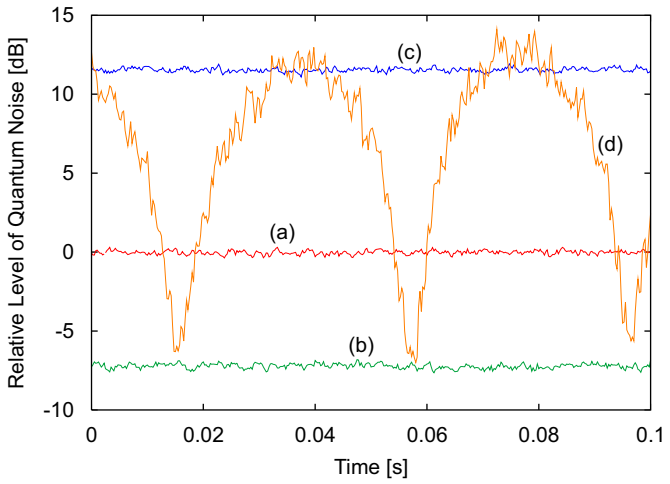


Fig. 23. −7.2 dB squeezing [24]. Power levels of quantum noise. (a) shot noise level. (b) LO phase is locked at the squeezed quadrature. (c) LO phase is locked at the anti-squeezed quadrature. (d) LO phase is scanned. These are normalized to make the shot noise level 0 dB. All traces except for (d) are averaged for 20 times.

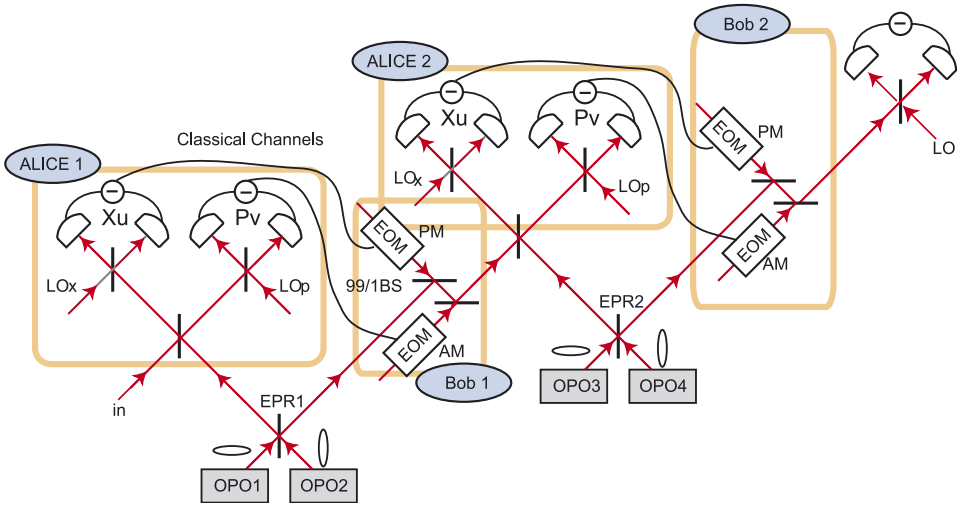


Fig. 24. Experimental setup for two cascaded quantum teleportation [43].

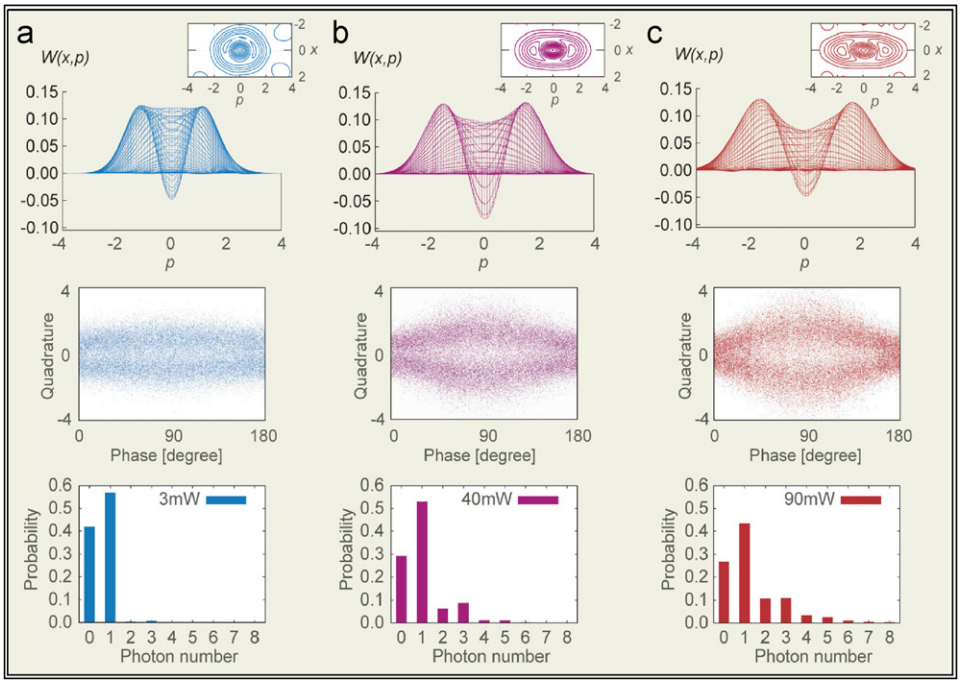


Fig. 25. Creation of non-Gaussian states which is called *Schrödinger kitten* and a single photon state with photon subtraction through photon number measurement of a squeezed vacuum [50]. $W(x, p)$ denotes the Wigner function, 'Quadrature' denotes the quadrature-phase amplitude, and 'Probability' denotes the photon number distribution.

From above consideration, the cubic phase gate can be realized with four cascaded CV quantum teleportation and photon number measurement. The teleportation fidelity for coherent state inputs for n cascaded CV quantum teleportation is following:

$$F_{\mathrm{coh}}(n) = \frac{1}{1 + ne^{-2r}}, \tag{31}$$

which is easily derived from Eq. (12), where r is a squeezing parameter of each squeezed vacuum for a resource of quantum entanglement. Since the success criterion for n cascaded CV quantum teleportation is $F_{\mathrm{coh}}(n) > 1/2$, we need at least -6 dB of squeezing ($e^{-2r} = 1/4$) for success of four cascaded CV quantum teleportation. Nowadays more squeezing than -6 dB is reported [24,25] (Fig. 23) and the realization is really promising. This line of experiment, two cascaded CV quantum teleportation, is reported by Yonezawa and Furusawa and the experimental setup is shown in Fig. 24 [43].

In addition to the cascaded quantum teleportation experiments, some groups are working on photon number measurement for non-Gaussian state preparation [44–46]. One of the examples is shown in Fig. 25 [50]. For combining this technique with Gaussian operations, especially with CV quantum teleportation experiments, we have to broaden the bandwidth of the experiments, because photon number measurements always see the whole bandwidth of the EPR beams in contrast with usual CV quantum teleportation experiments which use some particular sidebands. For the direction of this research, creation of broad band or time-domain EPR beams is reported and the results are shown in Fig. 3 [5]. The time resolution is determined by the bandwidth of the EPR beams.

Now we have enough bandwidth of the EPR beams, the next step is quantum teleportation of non-Gaussian states and then squeezing of a single photon state to create a *Schrödinger kitten* with a universal squeezer, where a *Schrödinger kitten* is superposition of coherent states as shown in the next section.

In any case, the cubic phase gate is one of the most promising non-Gaussian operation gates. By using the gate, we could accomplish the most fundamental non-Gaussian operation, i.e., entanglement distillation, toward universal CV quantum information processing.

5. Future prospects

Up to here, we have showed the basics and the recent progress on CV quantum teleportation and related quantum information processing. We believe that we will be able to make universal CV quantum information processing in the future with the presented techniques here. Moreover, since CV quantum information is higher dimensional version of qubits, the success of universal CV quantum information processing means the success of universal qubit quantum information processing. For example, we can convert CV quantum information to qubits as follows:

$$|0\rangle = c(|\alpha\rangle + |-\alpha\rangle),$$
$$|1\rangle = c(|\alpha\rangle - |-\alpha\rangle), \tag{32}$$

where the important thing here is that $|0\rangle$ and $|1\rangle$ are orthonormal. Recently, creation of $c(|\alpha\rangle - |-\alpha\rangle)$, so-called *Schrödinger kitten*, with photon subtraction technique is reported [47–50] and the example is shown in Fig. 25. By using this type of qubits, we also will be able to realize universal qubit quantum information processing.

As mentioned everywhere in this review, the most fundamental quantum information processing is quantum teleportation. So we should pursuit to improve the performance.

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